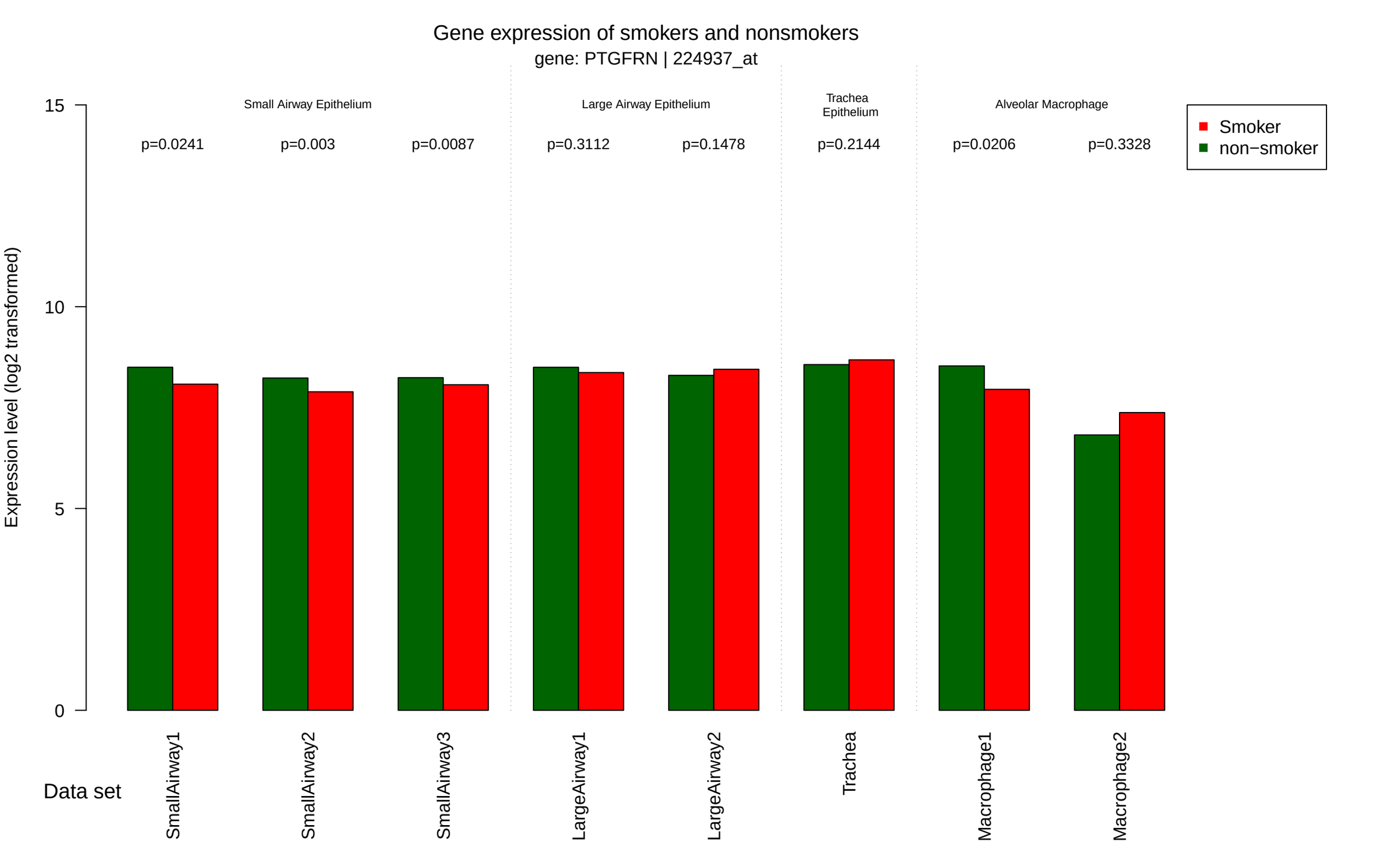
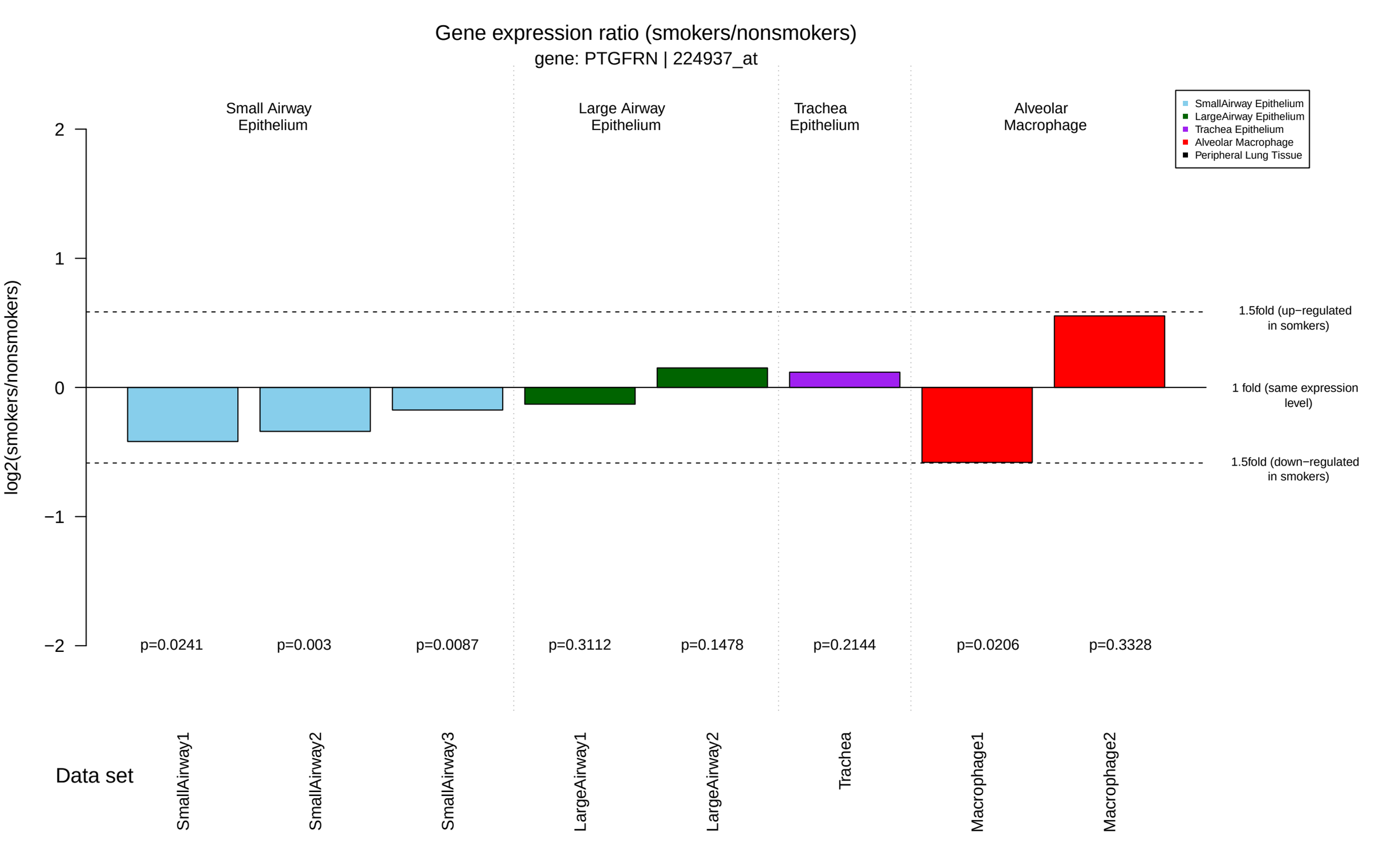
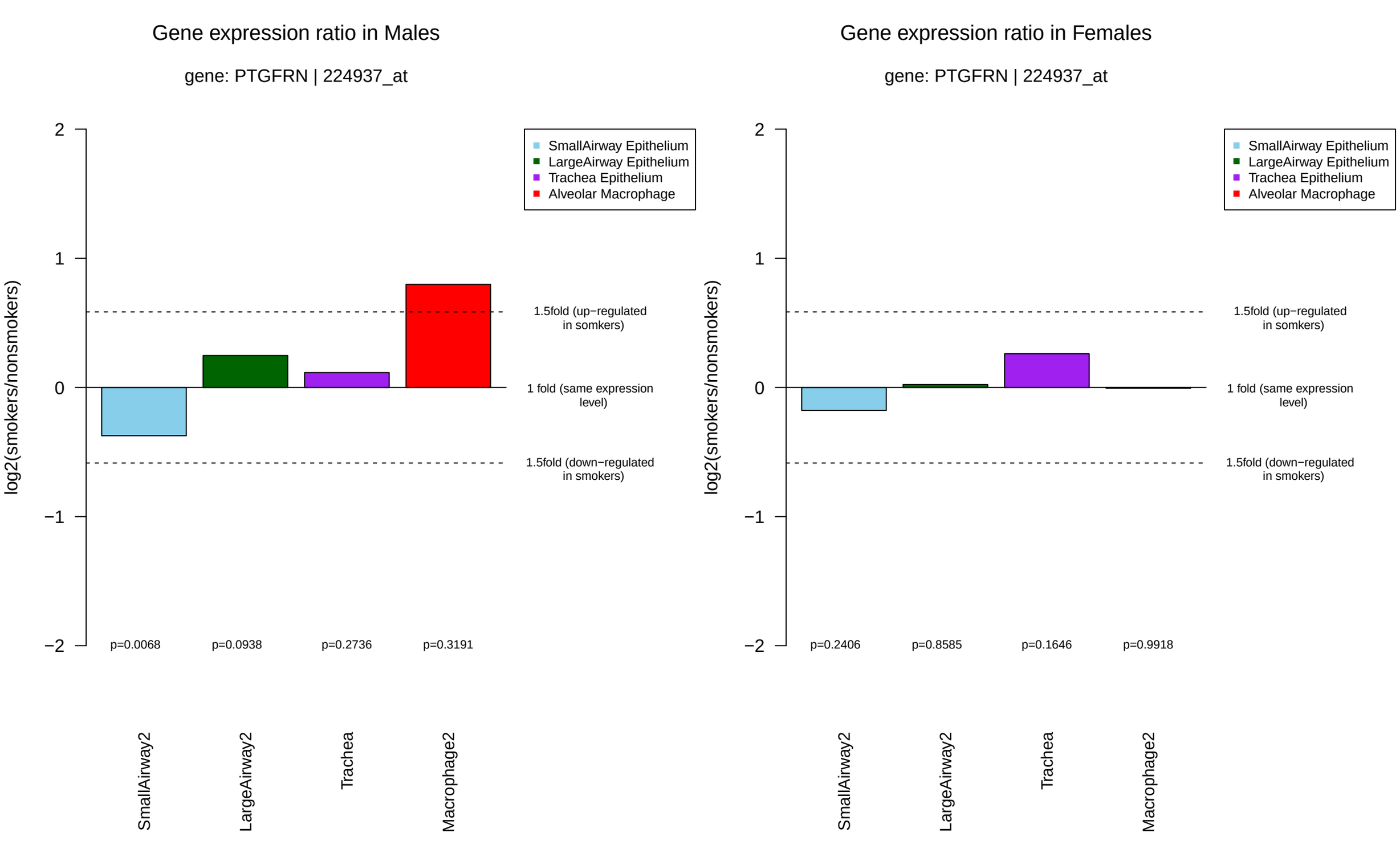
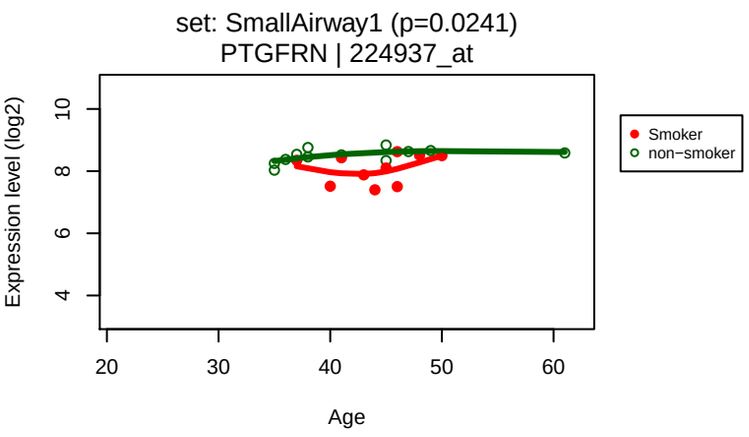
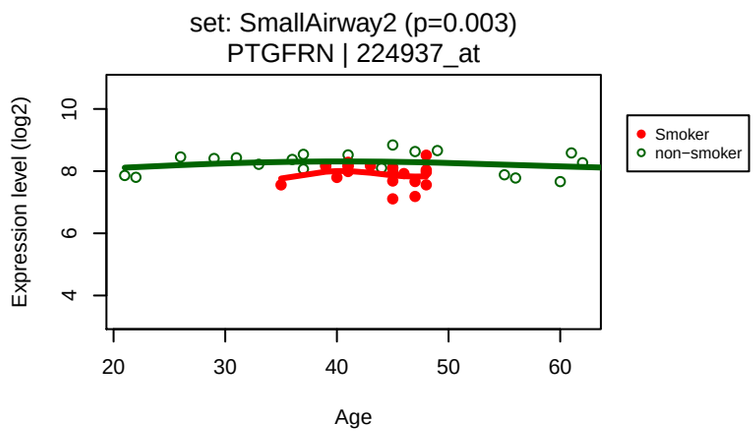
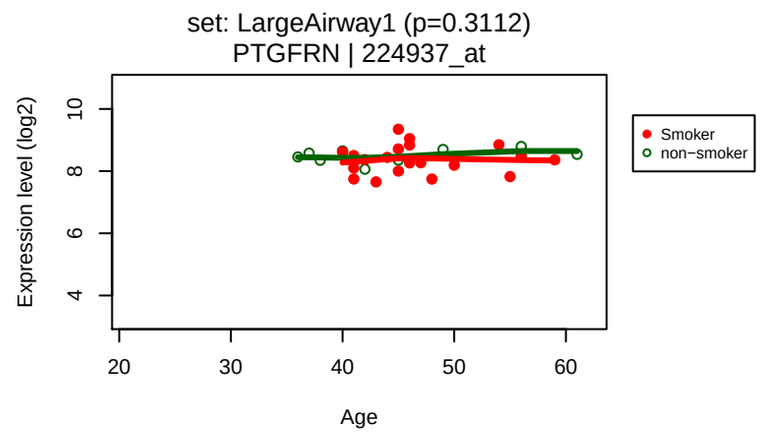

Gene expression of smokers and nonsmokers
gene: PTGFRN | 224937_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0241
p=0.003
p=0.0087
p=0.3112
p=0.1478
p=0.2144
p=0.0206
p=0.3328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

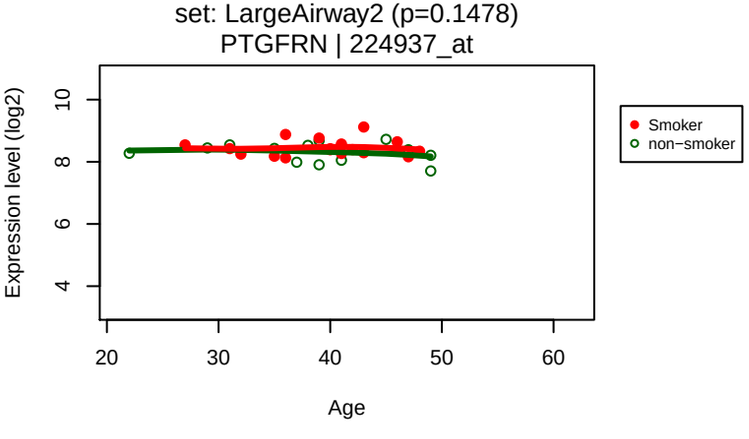

Gene expression ratio (smokers/nonsmokers)
gene: PTGFRN | 224937_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0241
p=0.003
p=0.0087
p=0.3112
p=0.1478
p=0.2144
p=0.0206
p=0.3328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

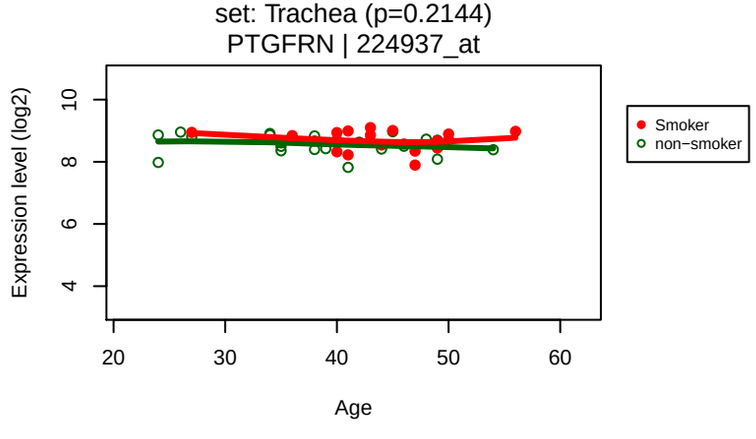

Gene expression ratio in Males
gene: PTGFRN | 224937_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0068
p=0.0938
p=0.2736
p=0.3191
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGFRN | 224937_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2406
p=0.8585
p=0.1646
p=0.9918
SmallAirway2
LargeAirway2
Trachea
Macrophage2

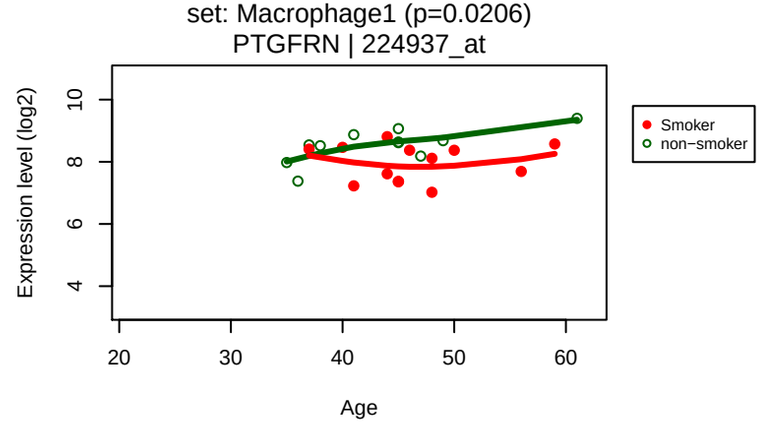

set: SmallAirway1 (p=0.0241)
PTGFRN | 224937_at
Expression level (log2)
Age
Smoker
non−smoker

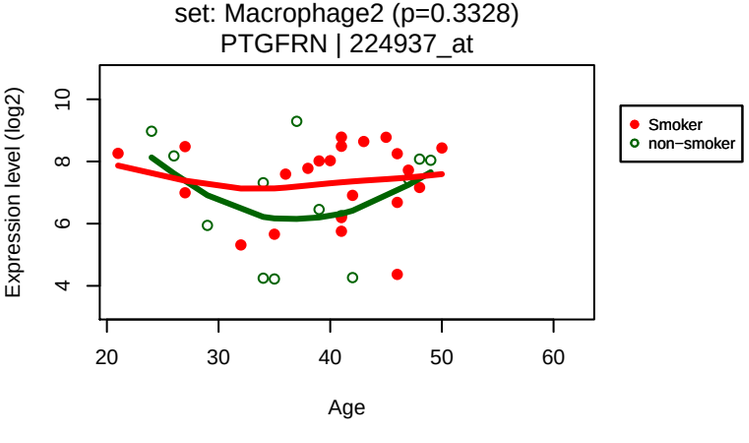

set: SmallAirway2 (p=0.003)
PTGFRN | 224937_at
Expression level (log2)
Age
Smoker
non−smoker

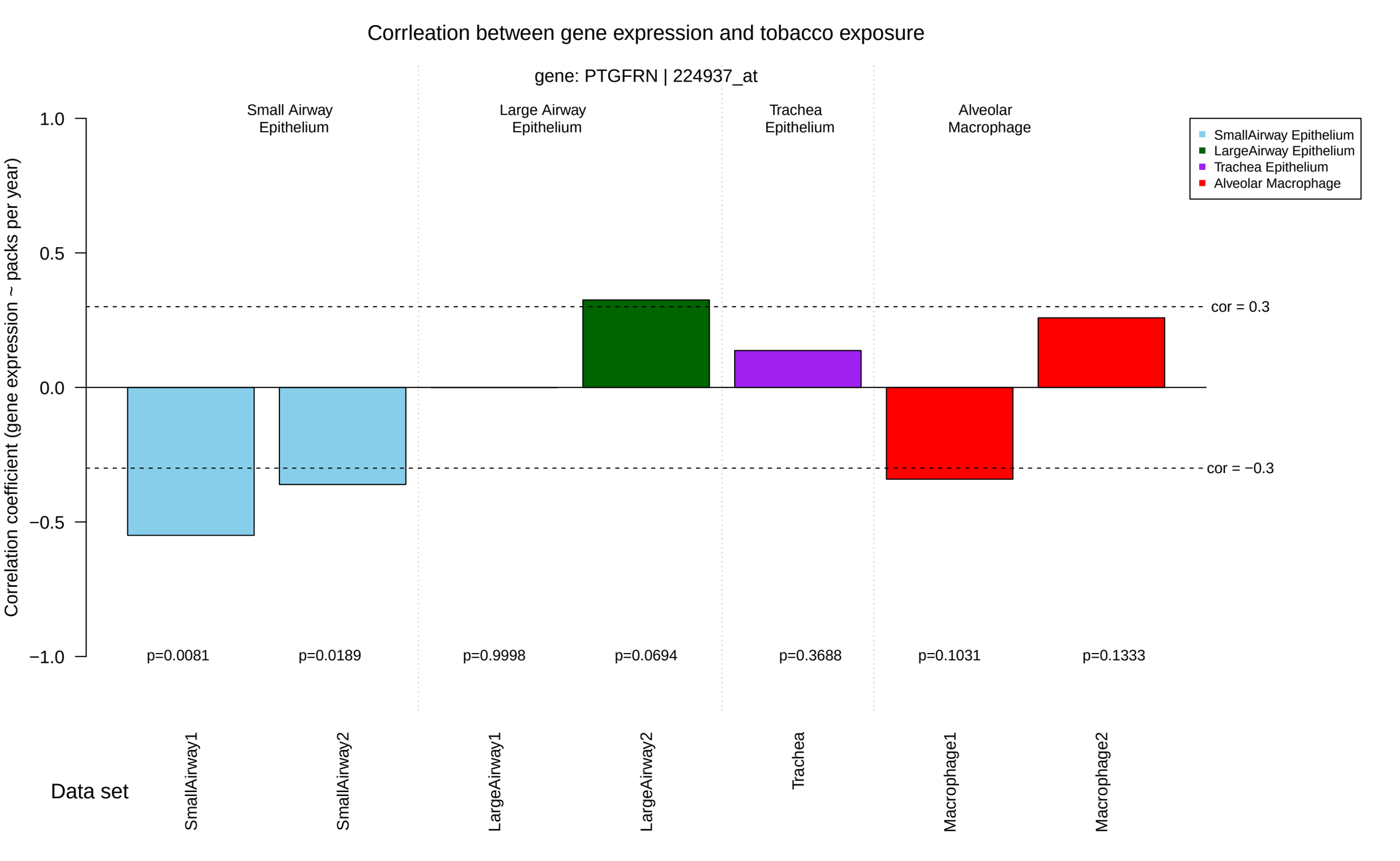

set: LargeAirway1 (p=0.3112)
PTGFRN | 224937_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1478)
PTGFRN | 224937_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2144)
PTGFRN | 224937_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0206)
PTGFRN | 224937_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3328)
PTGFRN | 224937_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGFRN | 224937_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0081
p=0.0189
p=0.9998
p=0.0694
p=0.3688
p=0.1031
p=0.1333
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2